黃 東 浩

+1 (571) XXXX-XXXX | donghao.huang@gmail.com | 美國弗吉尼亞州阿靈頓

新加坡公民 | 專利：10 項已授權，50+項審批中

職業概述

擁有 26 年以上技術與管理經驗的科技高管，覆蓋軟體工程、Web3、AI/ML/GenAI 以及支付創新領域。現任萬事達卡（Mastercard）新興技術方向（AI/Web3）全球研發負責人，同時攻讀生成式 AI 工程博士。已發表 24 項研究成果，涵蓋大語言模型（LLM）部署、LLM/ML 模型最佳化與應用，以及多智慧體系統。具備將學術研究轉化為金融服務領域企業級落地成果的成熟實踐與卓越業績。

教育背景

生成式人工智慧工程博士 (Doctor of Engineering) 2023 年 8 月 – 2026 年 5 月

新加坡管理大學，新加坡（在讀）

- 2025 年獲新加坡資訊通訊媒體發展局（IMDA）數字獎學金（研究生）
- 研究方向：檢索增強生成（RAG）、邊緣 AI、LLM 和 ML 模型最佳化、多智慧體系統、情感分析、推理型 LLM
- 在 AI 學術會議和期刊發表 24 篇同行評審論文（AAAI、AAMAS、NAACL、IJCNN、PAKDD、IEEE CAI、IEEE SSCI、IEEE ICDM、IEEE Intelligent Systems、Artificial Intelligence Reviews）

軟體工程技術碩士 2004 年 1 月 – 2008 年 1 月

新加坡國立大學，新加坡

新加坡經濟發展局(EDB)研究與培訓專案證書 2001 年 9 月 – 2002 年 8 月

新加坡國立大學光電子中心，新加坡

理論物理學碩士 1998 年 9 月 – 2001 年 7 月

北京大學，中國北京

- 在物理學術期刊發表 2 篇同行評審論文

工程物理學學士 1989 年 9 月 – 1995 年 7 月

清華大學，中國北京

職業經歷

研發副總裁 / 全球新興技術負責人 – Mastercard（美國） 2025 年 8 月 – 至今

美國弗吉尼亞州阿靈頓

- 領導 AI 新興技術（EmTech）領域的全球研發計劃
- 領導 5 個戰略性"大賭注"專案的 EmTech 研究——旨在塑造 Mastercard 下一代支付基礎設施和 AI 驅動商業解決方案的高影響力機密專案
- 組織並主持 SENTIRE'25（第 15 屆情感分析與語言關聯資料研討會），與 IEEE ICDM 2025 聯合舉辦，2025 年 11 月 13 日，美國華盛頓特區

研發副總裁 / 全球新興技術負責人 – Mastercard 亞太 2022 年 5 月 – 2025 年 7 月

新加坡

- 領導新興技術（EmTech）領域的全球研發計劃：AI / 機器學習 / 生成式 AI 和 Web3
- AI/LLM/GenAI 專案：
  - 領導 Mastercard 助手平臺（MAP）的 EmTech 研究，這是一套旨在簡化 Mastercard 產品客戶入門流程的數字助手
    - 使用 LangChain、FAISS 向量資料庫和 HuggingFace 嵌入構建初始 RAG 原型
    - 在 NVIDIA A40 和 L40 GPU 上對開源 LLM（Llama 2、Orca 2）進行早期基準測試，展示 24-49%的推理速度提升
    - 開發創新的重複感知效能（RAP）評估框架，現已部署於 MAP 生產環境
    - 領導使用 Ollama 在多種硬體平臺上進行邊緣感知 LLM 部署研究
  - 使用 LLM（GPT-4/5、Claude、Llama、Qwen）開創企業 GenAI 應用，實現支付/商業自動化
    - 使用 QLoRA 在多 GPU 配置上微調 70B+引數模型，用於中英翻譯和邏輯推理
    - 開發多智慧體系統，使用邊緣部署 LLM 實現 99.9%準確率的發票對賬
    - 開發 HMASP——首個實現端到端智慧體支付工作流的基於 LLM 的多智慧體系統
    - 開發 CMAS4G2 協作多智慧體系統，用於自動生成 Gherkin 驗收測試
    - 建立 LLM-as-a-Judge 評估框架，相比 GPT-5 實現 47 倍成本降低
- Web3 專案：
  - 開發 Mastercard 代幣化卡片 NFT 解決方案，支援將支付卡作為靈魂繫結代幣（SBT）進行代幣化
  - 開發 DigiPay，利用 Layer 2 區塊鏈和 ERC-4337 賬戶抽象的 Web3 微支付穩定幣解決方案

研發副總裁 / 新加坡研發負責人 – Mastercard 亞太 2011 年 8 月 – 2022 年 5 月

新加坡

- 建立並領導新加坡研發團隊——Mastercard 全球創新與研發部門；管理位於新加坡的高績效研發團隊
- 從事支付/商業領域的創新與研發，涵蓋前沿技術：EMV 晶片、物聯網、二維碼支付、車聯網、人形機器人、對話式 AI、AR/VR、區塊鏈和數字資產、Web3、NFT、DeFi、元宇宙
- 推動複雜專案從創意到商業化交接的全流程；促進專案向 Mastercard 商業化流程的過渡

- 作為技術創新的倡導者，與各類合作伙伴（內部部門、外部合作伙伴、大學）協作，確保Mastercard 在支付領域建立技術卓越的聲譽
- 主要商業化專案：
  - 2021 年 11 月：Mastercard 與新奧爾良市和 MoCaFi 合作推出 Crescent City Card 專案
  - 2020 年 9 月：Mastercard 與三星、Airtel 和 Asante 合作，透過按需付費服務推動非洲數字普惠
  - 2018 年 10 月：Mastercard、Centenary Bank 和 M-KOPA Solar 在烏幹達推出首創能源解決方案
  - 2018 年 2 月：Mastercard 使用 Facebook Messenger 幫助小企業數字化
  - 2017 年 9 月：SPH Buzz 與 Mastercard 合作推出新型混合商店概念
  - 2017 年 2 月：BharatQR——全球首個可互操作的二維碼受理解決方案在印度推出
  - 2016 年 5 月：MasterCard 為軟銀機器人人形機器人"Pepper"提供首個商業應用
  - 2015 年 9 月：MasterCard 變革援助物資分發方式
  - 2012 年 8 月：StarHub 與 DBS、EZ-Link 和 MasterCard 合作推出智慧錢包

高階軟體工程師 – PayPal 2008 年 7 月 – 2011 年 8 月

新加坡

- 領導軟體開發團隊，負責一系列針對巴西市場的專案，以滿足這一快速增長市場的需求
- 限額/驗證/合規領域的主題專家
- 主要商業化專案：
  - 領導 12 人開發團隊，在 8 個月內交付最大的歐盟合規專案（EU KYB Receiving Limit）
  - 構建中國銀聯（CUP）整合，實現實時交易處理

軟體研發經理 – Streaming21（新加坡） 2007 年 10 月 – 2008 年 7 月

新加坡

- 建立研發中心並領導媒體中繼伺服器的開發——一款領先的 IPTV 流媒體伺服器

軟體開發負責人 – Schmidt Electronics (S.E.A.) 2006 年 1 月 – 2007 年 9 月

新加坡

軟體工程師 – Unaxis 新加坡 2005 年 2 月 – 2005 年 12 月

新加坡

軟體工程師 – ASM Technology 新加坡 2002 年 9 月 – 2005 年 2 月

新加坡

軟體工程師 – 北京智達科技 1995 年 8 月 – 1998 年 8 月

中國北京

志願服務經歷

會長 – 寶嚴（新加坡）教育協會	2025 年 11 月 – 至今
副會長 – 北京大學新加坡校友會	2016 年 6 月 – 2020 年 6 月
導師 – Startupbootcamp 金融科技新加坡	2015 年 1 月 – 2019 年 1 月
IT 與支付顧問 – ConneXions 國際（新加坡）	2011 年 3 月 – 2015 年 12 月
技術顧問 – 聖安德烈自閉症中心（新加坡）	2006 年 11 月 – 2008 年 7 月

技術技能

- AI/ML/GenAI：LLM（GPT-4/5、Claude、Gemini、Llama、Qwen、DeepSeek-R1、Mistral、Magistral、Phi、Nemotron）、PyTorch、Transformers、PEFT、LoRA/QLoRA 微調、LlamaFactory、Unsloth、RAG、LangChain、LangGraph、Autogen、Ollama、LM Studio、FAISS、向量資料庫、少樣本學習、微調、提示工程、情感分析、NLP
- GPU/硬體：NVIDIA H100、L40、A40、RTX A6000、RTX 4080、RTX 4090、Jetson AGX Orin、DGX Spark；Apple M 系列；4 位量化、模型最佳化、邊緣 AI 部署
- 程式語言與框架：Python、C/C++、Java、NodeJS、React、Vue、Spring Boot/Cloud、Solidity
- 雲與基礎設施：AWS、Azure、Heroku；微服務、CI/CD、Docker
- Web3/區塊鏈：區塊鏈、IPFS、NFT、DeFi、x402 協議

專利

（10 項已授權，50+項待審 — 均歸屬萬事達卡）

已授權專利

- **US12417452B2** – Smart chip payment acceptance (2025)
- **US12217246B2** – Method and system for use of an EMV card in a multi-signature wallet for cryptocurrency transactions (2025)
- **US11715100B2** – Electronic system and computerized method for verification of transacting parties to process transactions (2023)
- **US11615406B2** – Method and system for providing a service at a self-service machine (2023)
- **US10535034B2** – Item delivery management systems and methods (2020)
- **US10482499B2** – Method for conducting a transaction (2019)
- **US10423949B2** – Vending machine transactions (2019)
- **US10083427B2** – Method for receiving an electronic receipt of an electronic payment transaction into a mobile device (2018)
- **US9836729B2** – Method and system for conducting a payment transaction and corresponding devices (2017)
- **US8712914B2** – Method and system for facilitating micropayments in a financial transaction system (2014)

部分待審申請

- **WO2026063867A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **US20260080407A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **WO2024058723A1** – Digitization of payment cards for web 3.0 and metaverse transactions (2022)

- **US20230059546A1** – Access control system (2021)
- **EP4356332A2** – Method and system for mediated cross ledger stable coin atomic swaps using hashlocks (2021)
- **US20210406887A1** – Method and system for merchant acceptance of cryptocurrency via payment rails (2020)
- **US20210158316A1** – Electronic system and computerized method for controlling operation of service devices (2019)
- **US20200175489A1** – Method and system for QR code originated vending (2017)
- **WO2017200643A1** – Method and system for allocating and transacting with multiple non-financial commodities (2016)
- **US20180025332A1** – Transaction facilitation (2016)

... 以及 40+項其他待審專利申請，涵蓋支付技術、區塊鏈、AI 和物聯網領域。

精選論文

1. Chua, J., 等. (2026). 一種新型基於大語言模型的層級多智慧體支付系統. PAKDD 2026.
2. Huang, D., 等. (2026). 任務複雜度的重要性：大語言模型情感分析推理的實證研究. PAKDD 2026.
3. Huang, D., & Wang, Z. (2025). 使用 DeepSeek-R1 的可解釋情感分析. IEEE Intelligent Systems.
4. Huang, D., 等. (2025). 邊緣 LLM：效能與效率評估. IJCNN 2025.
5. Huang, D., 等. (2025). RAP：平衡重複與效能的指標. NAACL 2025.
6. Wang, Z., Huang, D., 等. (2025). 中文情感分析綜述：主題、方法與趨勢. Artificial Intelligence Review.
7. Lin, W. B., Huang, D. H., 等. (2001). 弱相互作用大質量粒子的非熱產生與亞星系結構. Physical Review Letters.

... 及其它 19 篇同行評審論文（人工智慧領域共 24 篇，物理領域共 2 篇）

DONGHAO HUANG

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Singapore Citizen | Patents: 10 granted, 50+ pending

PROFESSIONAL SUMMARY

Technology executive with 26+ years of experience across software engineering, Web3, AI/ML/GenAI, and payments innovation. Currently leading global R&D for Mastercard Emerging Technology (AI/Web3) while pursuing a Doctor of Engineering in Generative AI. Author of 24 published research works spanning LLM deployment, LLM/ML model optimization and applications, and multi-agent systems. Proven track record of translating academic research into enterprise-scale financial services solutions.

EDUCATION

Aug 2023 – May 2026 (ongoing) Singapore Management University Singapore
Doctor of Engineering (EngD) in Generative AI

- Awarded IMDA SG Digital Scholarship (Postgraduate) 2025
- Research focus: Retrieval-Augmented Generation (RAG), Edge AI, LLM and ML model optimization, multi-agent systems, sentiment analysis, reasoning LLMs
- Published 24 peer-reviewed papers at AI conferences and journals (AAMAS, AAAI, NAACL, IJCNN, PAKDD, IEEE CAI, IEEE SSCI, IEEE ICDM, IEEE Intelligent Systems, Artificial Intelligence Reviews)

Jan 2004 – Jan 2008 National University of Singapore Singapore
Master of Technology in Software Engineering

Sep 2001 – Aug 2002 Centre for Optoelectronics, NUS Singapore
Certificate, EDB Research and Training Programme

Sep 1998 – Jul 2001 Peking University Beijing, China
Master of Science in Theoretical Physics

- Published 2 peer-reviewed papers in physics journals

Sep 1989 – Jul 1995 Tsinghua University Beijing, China
Bachelor of Engineering in Engineering Physics

PROFESSIONAL EXPERIENCE

Aug 2025 – to date Mastercard (U.S.) Arlington, VA, USA
VP, R&D / Global EmTech Lead

- Lead global R&D initiatives in AI Emerging Technology (EmTech) domains
- Lead EmTech research for 5 strategic "Big Bets" projects – high-impact, confidential initiatives aimed at shaping Mastercard's next-generation payment infrastructure and AI-powered commerce solutions
- Organized and hosted SENTIRE'25 (15th Workshop on Sentiment Analysis and Linguistic Linked Data) co-located with IEEE ICDM 2025, November 13th, Washington DC.

May 2022 – Jul 2025 Mastercard Asia/Pacific Pte Ltd Singapore

VP, R&D / Global EmTech Lead

- Lead global R&D initiatives in Emerging Technology (EmTech) domains: AI / Machine Learning / Generative AI and Web3.
- **AI/LLM/GenAI Initiatives:**
  - Led EmTech research for Mastercard Assistant Platform (MAP)**,** a suite of digital assistants designed to simplify the customer onboarding process for Mastercard products
    - ➢ Built the initial Retrieval-Augmented Generation (RAG) prototype using LangChain, FAISS vector databases, and HuggingFace embeddings to power enterprise-grade knowledge systems.
    - ➢ Conducted early benchmarking of open-source LLMs (Llama 2, Orca 2) on NVIDIA A40 and L40 GPUs (48GB each), demonstrating 24-49% inference speed improvement on L40 vs A40.
    - ➢ Developed a novel Repetition-aware Performance (RAP) evaluation framework for LLM quality assessment, now fully deployed in the MAP production environment.
    - ➢ Led research on edge-aware LLM deployment using the Ollama framework across diverse hardware platforms (NVIDIA RTX A6000, NVIDIA RTX 4090, NVIDIA Jetson AGX Orin, Apple M-series—directly informing MAP's deployment strategy.
  - Pioneered enterprise GenAI applications using LLMs (GPT-4/5, Claude, Llama, Qwen) for payments/commerce automation
    - ➢ Fine-tuned 70B+ parameter models (Llama-3.1-70B, Qwen2-72B) using QLoRA on multi-GPU setups (4× NVIDIA L40 GPUs) for Chinese-English translation and logical reasoning
    - ➢ Developed multi-agent systems for invoice reconciliation achieving 99.9% accuracy using edge-deployed LLMs (qwen2.5:32b)
    - ➢ Developed hierarchy multi-agent systems for conversational payment agent (HMASP) - first LLM-based multi-agent system to implement end-to-end agentic payment workflows, enabling external AI agents to converse and complete payments using natural language with minimal integration overhead
    - ➢ Developed CMAS4G2, a cooperative multi-agent system for automated Gherkin acceptance test generation using Microsoft's AutoGen Swarm framework; demonstrated open-weight models on consumer hardware (NVIDIA RTX 4090, 24GB VRAM) approach proprietary performance (GPT-OSS 20B: 87.6% success vs Claude Sonnet 4: 90.4%)
    - ➢ Created LLM-as-a-Judge (LAJ) evaluation framework achieving 47× cost reduction vs GPT-5 with negligible accuracy loss for automated test coverage assessment
    - ➢ Evaluated reasoning LLMs (DeepSeek-R1, Qwen3, GPT-OSS) for sentiment analysis with cross-platform performance benchmarking on NVIDIA H100, NVIDIA GeForce RTX 4090, Apple M3 Max
    - ➢ Developed energy efficiency metrics (Energy per Inference, Performance-to-Power Ratio) for sustainable AI deployment
- **Web3 Initiatives:**
  - Developed Mastercard Tokenized Card NFT solution, enabling seamless tokenization of payment cards as Soul Bound Tokens (SBTs) into Web3 wallets
    - ➢ Enables users to store credit cards securely as non-transferable NFTs, providing publicly verifiable digital identity for web3 wallet addresses

    - Integrates with crypto exchanges, merchant partners, and payment acquirers through smart contract architecture
    - Simplifies Web3 user onboarding by eliminating the need to switch between multiple applications and cumbersome digital wallet funding processes
  - Developed DigiPay, a Web3 micropayment stablecoin solution leveraging Layer 2 blockchain and ERC-4337 account abstraction
    - Implemented Digicents smart contract token with high TPS, low-latency blockchain for fast, efficient, and affordable transactions
    - Built smart contract wallet accounts with auto-top-up and auto-withdrawal features, eliminating manual on-ramp/off-ramp processes
    - Integrated with Tokenized Card NFT for secure funding and transaction authentication within the Digicents ecosystem

Aug 2011 – May 2022 Mastercard Asia/Pacific Pte Ltd Singapore

VP, R&D / Head of Singapore R&D

- Found and lead Singapore R&D team in Mastercard Foundry (previously known as Mastercard Labs) – Mastercard's global innovation and R&D division; managing a high performing R&D team based in Singapore.
- Work on innovation and R&D in payment/commerce domains across lots of latest and cutting-edge technologies, including but not limited to, EMV Chip technologies, Internet of Things, QR Payments, Connected Car, Humanoid Robots, Conversational AI, AR/VR, Blockchain and Digital Assets, Web3, NFT, DeFi, Metaverse and so on
- Drive complex initiatives from idea through to commercialization hand-over; facilitate the transition of initiatives into Mastercard commercialization process
- As a champion for Technological Innovation, collaborate with various partners (internal departments, external partners, universities) to ensure Mastercard attains a reputation for technological excellence in the Payments space
- **Key Commercialized Projects**
  - Nov 2021: Mastercard partners with city of New Orleans and MoCaFi to announce Crescent City Card program: https://www.mastercard.com/news/press/2021/november/mastercard-partners-with-city-of-new-orleans-and-mocafi-to-announce-crescent-city-card-program/
  - Sep 2020: Mastercard partners with Samsung, Airtel and Asante to drive digital inclusion in Africa through Pay-on-Demand services: https://newsroom.mastercard.com/mea/press-releases/mastercard-partners-with-samsung-airtel-and-asante-to-drive-digital-inclusion-in-africa-through-pay-on-demand-services/
  - Oct 2018: Mastercard, Centenary Bank and M-KOPA Solar roll-out first of its kind energy solution in Uganda: https://newsroom.mastercard.com/mea/press-releases/mastercard-centenary-bank-and-m-kopa-solar-roll-out-first-of-its-kind-energy-solution-in-uganda/
  - Feb 2018: Mastercard Uses Facebook Messenger To Help Small Businesses Go Digital: https://newsroom.mastercard.com/press-releases/mastercard-uses-facebook-messenger-help-small-businesses-go-digital/
  - Sep 2017: SPH Buzz Partners With Mastercard To Launch New Hybrid Store Concept: https://newsroom.mastercard.com/asia-pacific/press-releases/sph-buzz-partners-with-mastercard-to-launch-new-hybrid-store-concept/

- Feb 2017: BharatQR – World's first interoperable QR code acceptance solution launched in India: https://newsroom.mastercard.com/asia-pacific/press-releases/bharatqr-worlds-first-interoperable-qr-code-acceptance-solution-launched-in-india/
- May 2016: MasterCard Powers First Commerce Application within SoftBank Robotics' Humanoid Robot "Pepper": https://newsroom.mastercard.com/press-releases/mastercard-powers-first-commerce-application-within-softbank-robotics-humanoid-robot-pepper/
- Sep 2015: MasterCard Transforms Aid Distribution: https://newsroom.mastercard.com/press-releases/mastercard-transforms-aid-distribution/
- Aug 2012: StarHub unveils 'SmartWallet', in collaboration with DBS, EZ-Link and MasterCard: https://www.starhub.com/about-us/newsroom/2012/august/starhub-unveils-smartwallet--in-collaboration-with-dbs--ez-link-.html

Jul 2008 – Aug 2011 PayPal Singapore Development Centre Singapore
Senior Software Engineer / Team Lead

- Lead a team of software developers, working on a pipeline of projects related to Brazil to cater to this fast growing market
- Subject Matter Expert in limits / verification / compliance domain
- **Key Commercialized Projects**
  - Led 12-developer team delivering largest EU compliance project (EU KYB Receiving Limit) in 8 months
  - Built China UnionPay (CUP) integration enabling real-time transaction processing

Oct 2007 – Jul 2008 Streaming21 (Singapore) Pte Ltd Singapore
Software R&D Manager

- Established an R&D centre and led development of Media Relay Server, a leading IPTV streaming server.

Jan 2006 – Sep 2007 Schmidt Electronics (S.E.A.) Pte Ltd Singapore
Leader, Software Development

Feb 2005 – Dec 2005 Unaxis Singapore Pte Ltd Singapore
Software Engineer

Sep 2002 – Feb 2005 ASM Technology Singapore Pte Ltd Singapore
Software Engineer

Aug 1995 – Aug 1998 Beijing Zhida Technology Pte Ltd Beijing, China
Software Engineer

VOLUNTEERING EXPERIENCE

Nov 2025 – to date Bao Yan (Singapore) Education Society
President

Jun 2016 – Jun 2020 Peking University Alumni Association (Singapore)
Vice President

Jan 2015 – Jan 2019 Startupbootcamp FinTech Singapore
Mentor

Mar 2011 – Dec 2015 ConneXions International Singapore
IT and Payment Consultant

Nov 2006 – Jul 2008 St. Andrew's Autism Centre Singapore
Technology Consultant

TECHNICAL SKILLS

- **AI/ML/GenAI:** LLMs (GPT-4, GPT-5, Claude, Gemini, Llama, Qwen, DeepSeek-R1, Mistral, Magistral, Phi, Nemotron), PyTorch, Transformers, LLM/ML Model Fine-tuning and Optimization, PEFT, LoRA/QLoRA Fine-tuning, LlamaFactory, Unsloth, RAG, LangChain, LangGraph, Autogen, Ollama, LM Studio, Vector Databases, FAISS, Chroma, Prompt Engineering, Few-shot Learning, NLP, Sentiment Analysis, Agentic AI, OpenClaw, NemoClaw, Edge AI Deployment, Energy-efficient Inference
- **GPU/Hardware:** NVIDIA H100, L40, A40, RTX 4080, RTX 4090, RTX A6000, Jetson AGX Orin, DGX Spark; Apple M-series
- **Languages & Frameworks:** Python, C/C++, Java, NodeJS, React, Vue, Spring Boot/Cloud, Solidity
- **Cloud & Infrastructure:** AWS, Azure, Heroku; Microservices, CI/CD, Docker
- **Web3/Blockchain:** Blockchain, Smart Contract, IPFS, NFT, DeFi

PATENTS

(10 granted, 50+ pending — all assigned to Mastercard)

Granted Patents

- **US12417452B2** – Smart chip payment acceptance (2025)
- **US12217246B2** – Method and system for use of an EMV card in a multi-signature wallet for cryptocurrency transactions (2025)
- **US11715100B2** – Electronic system and computerized method for verification of transacting parties to process transactions (2023)
- **US11615406B2** – Method and system for providing a service at a self-service machine (2023)
- **US10535034B2** – Item delivery management systems and methods (2020)
- **US10482499B2** – Method for conducting a transaction (2019)
- **US10423949B2** – Vending machine transactions (2019)
- **US10083427B2** – Method for receiving an electronic receipt of an electronic payment transaction into a mobile device (2018)
- **US9836729B2** – Method and system for conducting a payment transaction and corresponding devices (2017)
- **US8712914B2** – Method and system for facilitating micropayments in a financial transaction system (2014)

Selected Pending Applications

- **WO2026063867A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **US20260080407A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **WO2024058723A1** – Digitization of payment cards for web 3.0 and metaverse transactions (2022)
- **US20230059546A1** – Access control system (2021)
- **EP4356332A2** – Method and system for mediated cross ledger stable coin atomic swaps using hashlocks (2021)
- **US20210406887A1** – Method and system for merchant acceptance of cryptocurrency via payment rails (2020)

- **US20210158316A1** – Electronic system and computerized method for controlling operation of service devices (2019)
- **US20200175489A1** – Method and system for QR code originated vending (2017)
- **WO2017200643A1** – Method and system for allocating and transacting with multiple non-financial commodities (2016)
- **US20180025332A1** – Transaction facilitation (2016)

… *and 40+ additional pending patent applications across payment technology, blockchain, AI, and IoT domains.*

RESEARCH PAPERS

(papers at top-tier venues are highlighted in yellow; full texts are accessible via the hyperlinked paper titles)

Artificial Intelligence

1. Chua, J., Huang, D. & Wang, Z. (2026). A Novel Hierarchical Multi-Agent System for Payments Using LLMs. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026).*
2. Huang, D., & Wang, Z. (2026). Task Complexity Matters: An Empirical Study of Reasoning in LLMs for Sentiment Analysis. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026).*
3. Huang, D., Chua, J. K., & Wang, Z. (2026). Beyond Task Success: Measuring Workflow Fidelity in LLM-Based Agentic Payment Systems. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
4. Huang, D., Pai, P., & Wang, Z. (2026). Entity Matching with LLMs at Scale: Accuracy, Cost, and Reasoning Effects. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
5. Huang, D., Chew, S., & Wang, Z. (2026). CMAS4G2: A Locally Deployed Cooperative Multi-Agent System for Gherkin Acceptance Test Generation. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
6. Huang, D. & Pandey, M. (2026). Huang, D. & Pandey, M. (2026). An Agentic Voice-Based Assistant for Interactive Conversation and Guidance in Real-World Environments. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
7. Chua, J. K., Tse, C. H., Huang, D., & Wang, Z. (2026). ConvPayMAS: Conversational Payment Multi-Agent System with Agent-to-Agent Protocol and Three-Mandate Verification. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
8. Huang, D., Malwe, G., & Wang, Z. (2026). When Agents Fail to Act: A Diagnostic Framework for Tool Invocation Reliability in Multi-Agent LLM Systems. Accepted for publication in *2026 The 9th International Conference on Artificial Intelligence and Big Data (ICAIBD 2026).*
9. Huang, D., Chew S., Dutkiewicz A., & Wang, Z. (2026). LLM-as-a-Judge for Scalable Test Coverage Evaluation: Accuracy, Operational Reliability, and Cost. Presented at *AAAI 2026 Workshop on Next-Gen Code Development with Collaborative AI Agents.*
10. D. Huang, N. Teo, A. Chekuru, E. Deutschman and Z. Wang, "Deploying Reasoning LLMs for Sentiment Analysis: Architecture Trade-offs on Consumer Hardware," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2048-2055, doi: 10.1109/ICDMW69685.2025.00249.
11. S. Samuel, D. Huang, J. T. Y. Hong and Z. Wang, "Integrating LLM Sentiment Analysis into Machine Learning for Soccer Betting," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2156-2164, doi: 10.1109/ICDMW69685.2025.00262.
12. D. Huang and Z. Wang, "Explainable Sentiment Analysis With DeepSeek-R1: Performance, Efficiency, and Few-Shot Learning," in *IEEE Intelligent Systems*, vol. 40, no. 6, pp. 52-63, Nov.-Dec. 2025, doi: 10.1109/MIS.2025.3614967.

13. Huang, D., Ng, D., Pen, H., Wang, Z., & Cambria E. (2025). Evaluating the Impact of LLM-Manipulated Content on Fake News Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2025.* Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_28
14. Teo, N., Huang, D., Cambria, E., Wang, Z. (2025). Large Language Models for Logical Fallacy Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2025.* Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_29
15. D. Huang and Z. Wang, "LLMs at the Edge: Performance and Efficiency Evaluation with Ollama on Diverse Hardware," *2025 International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025, pp. 1-8, doi: 10.1109/IJCNN64981.2025.11228317.
16. Huang, D., Nguyen, T. S., Liausvia, F., & Wang, Z. (2025, April). RAP: A Metric for Balancing Repetition and Performance in Open-Source Large Language Models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 1479-1496).
17. Huang, D., Malwe, G., Varshney, R., Tse, A., & Wang, Z. (2025, March). Scalable Invoice Reconciliation for SMEs: Edge-Deployed LLMs in Multi-Agent Systems. In *2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)* (pp. 1-6). IEEE.
18. D. Huang and Z. Wang, "Optimizing Chinese-to-English Translation Using Large Language Models, " *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-7, doi: 10.1109/CI-NLPSoMe64976.2025.10970768.
19. D. Huang and Z. Wang, "Logical Reasoning with LLMs via Few-Shot Prompting and Fine-Tuning: A Case Study on Turtle Soup Puzzles," *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-5, doi: 10.1109/CI-NLPSoMeCompanion65206.2025.10977921.
20. Wang, Z., Huang, D., Cui, J. et al. (2025). A review of Chinese sentiment analysis: subjects, methods, and trends. Artif Intell Rev 58, 75 (2025).
21. Huang, D., Fu, X., Yin, X., Pen, H., & Wang, Z. (2024). Automating maritime risk data collection and identification leveraging large language models. In *2024 IEEE International Conference on Data Mining Workshops (ICDMW)* (pp. 433–439). IEEE.
22. Huang, D., Hu, Z., & Wang, Z. (2024, June). Performance Analysis of Llama 2 Among Other LLMs. In *2024 IEEE Conference on Artificial Intelligence (CAI)* (pp. 1081-1085). IEEE.
23. Huang, D., & Wang, Z. (2024, April). Evaluation of Orca 2 Against Other LLMs for Retrieval Augmented Generation. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_1
24. Huang, D., Samuel, S., Hyunh, Q. T., & Wang, Z. (2024, April). From Tweets to Token Sales: Assessing ICO Success Through Social Media Sentiments. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_5

Theoretical Physics

25. Lin, W. B., Huang, D. H., Zhang, X., & Brandenberger, R. (2001). Nonthermal production of weakly interacting massive particles and the subgalactic structure of the universe. *Physical Review Letters*, *86*(6), 954.
26. Huang, D. H., Lin, W. B., & Zhang, X. M. (2000). Remark on approximation in the calculation of the primordial spectrum generated during inflation. *Physical Review D*, *62*(8), 087302.